
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of vaccine development, disease elimination,
and remaining global challenges — 1796 to 2020

SOURCE

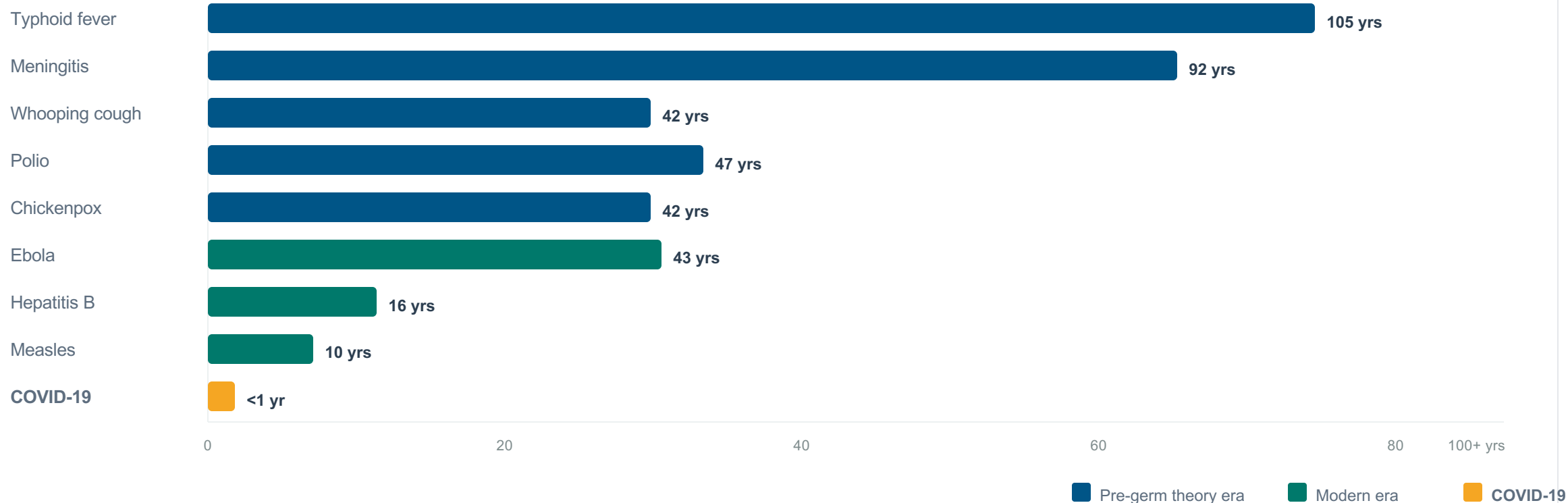
Our World in Data — Vaccination

DATE

April 2026

KEY INSIGHT

COVID-19 vaccine development took under 1 year — a >100x acceleration from typhoid fever's 105-year timeline.



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100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Impact

KEY INSIGHT

Four diseases achieved 100% elimination of both cases and deaths; pneumococcal disease lags at only 40.5% case reduction.

Disease	Pre-Vaccine Cases	Post-Vaccine	Case Reduction	Pre-Vaccine Deaths	Death Reduction	Status
Diphtheria	158 / million	0	100%	13.7 / million	100%	Eliminated
Measles	3,044 / million	0.2	99.99%	2.5 / million	100%	Near-zero
Smallpox	250 / million	0	100%	2.9 / million	100%	Eliminated
Acute polio	141 / million	0	100%	10 / million	100%	Eliminated
Pertussis	1,534 / million	52	96.6%	30.8 / million	99.7%	Controlled
Rubella	242 / million	0.04	99.98%	0.09 / million	100%	Eliminated
Hepatitis A	465 / million	51	89%	0.5 / million	88.7%	Incomplete
Acute hepatitis B	273 / million	44	83.9%	1 / million	83.6%	Incomplete
Pneumococcal	233 / million	139	40.5%	24 / million	31.3%	Gap remains

100% eliminated

Incomplete control

Major gap remains

KEY INSIGHT

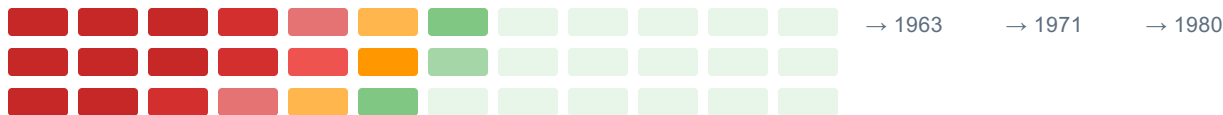
Before 1963, nearly every state had high measles incidence (100–1,000+ per 100,000). After the 1980 mandate, cases dropped to near zero.

Measles Incidence by State (1929–2022)

Cases per 100,000 population — conceptual heatmap representation



Representative state pattern:



Scale:  High (1000+) Moderate Near zero

Key Milestones**1963****First measles vaccine**

Developed by John Enders — marked the beginning of the measles decline.

1971**MMR vaccine introduced**

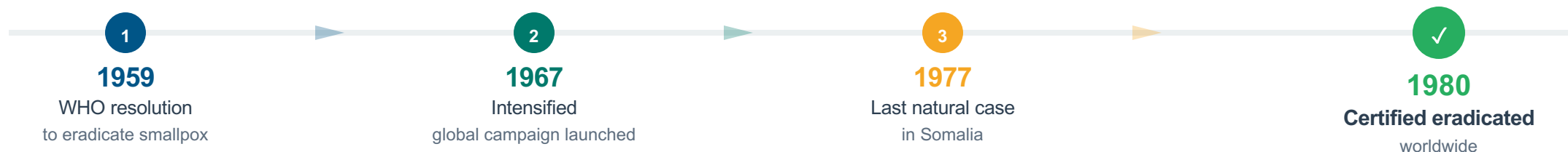
Developed by Maurice Hilleman — combined measles, mumps, and rubella protection.

1980**Mandatory kindergarten vaccination**

Cases dropped to near zero across all 50 states, with only sporadic outbreaks.

KEY INSIGHT

Smallpox was declared eradicated in 1980 — the only human disease fully eliminated through vaccination, after a 21-year global campaign.

The Road to Eradication**Why Smallpox Was Uniquely Suited for Eradication****Human-only infection**

The virus infected only humans, with no animal reservoir.

**Visible symptoms**

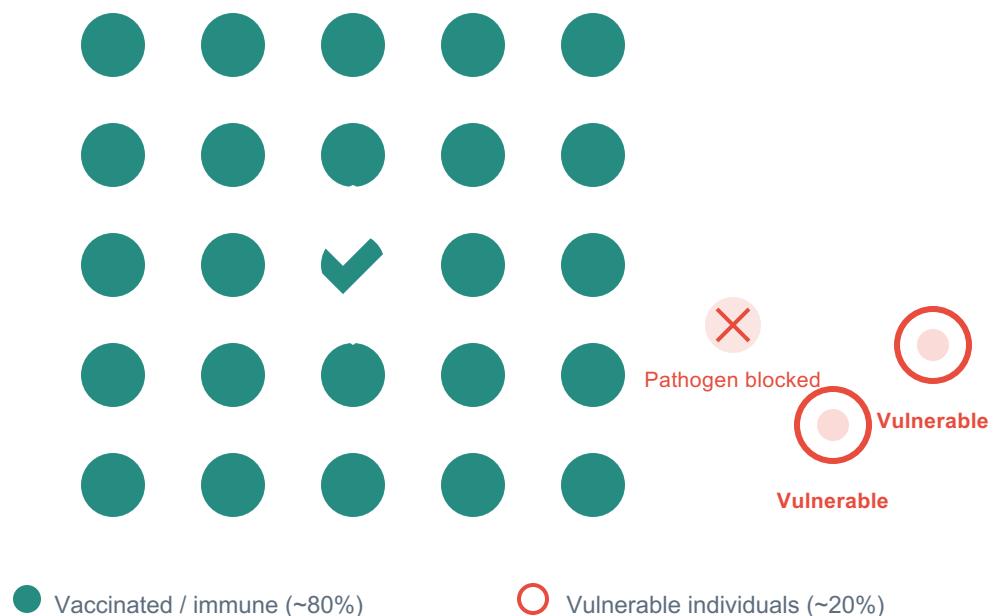
Symptoms were clearly distinguishable from other diseases.

**Cheap & effective vaccine**

The vaccine was affordable, stable, and highly effective.

KEY INSIGHT

When enough people are vaccinated, pathogens cannot spread effectively — shielding those who cannot be vaccinated.

Population-Level Protection**How Herd Immunity Works**

Vaccines protect not only individuals but entire communities. When a critical proportion of the population is immune, pathogens cannot find enough susceptible hosts to sustain transmission.

Those shielded include:

- Newborns too young for vaccines
- Immunocompromised individuals
- Cancer patients on chemotherapy

High vaccination coverage = community protection for the most vulnerable.

KEY INSIGHT

While some diseases are eliminated, pneumococcal disease, hepatitis A, and hepatitis B still show significant gaps in vaccine impact.



Pneumococcal disease

40.5%

Case reduction

31.3% death reduction



Hepatitis A

89%

Case reduction

88.7% death reduction



Acute hepatitis B

83.9%

Case reduction

83.6% death reduction

Why Gaps Remain



Access inequality

Not everyone has access to vaccines globally.



Supply shortages

Shortages put children at risk in low-resource settings.



Trust & understanding gaps

Poor scientific understanding can reduce trust and vaccine uptake.

KEY INSIGHT

Vaccines transformed public health — yet global health systems must prioritize equitable access, supply resilience, and public trust.

1. ELIMINATION

Vaccines achieved **100% elimination** of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.

2. ACCELERATION

Vaccine development timelines **collapsed from over 100 years** (typhoid) to **under 1 year** (COVID-19), driven by mRNA technology and genomic tools.

3. STATE-LEVEL IMPACT

Measles cases **dropped to near zero** across all 50 US states following the 1980 kindergarten vaccination mandate.

4. GLOBAL MILESTONE

Smallpox remains **the only human disease eradicated** through vaccination, certified in 1980 after a global campaign.

5. PERSISTING GAPS

Gaps persist for pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access, coverage, and newer vaccines.

CALL TO ACTION

Global health systems must prioritize **equitable access, supply chain resilience, and public trust** to finish the job.